

# **Sleep Deprivation and its Effect on the Academic Performances of Mahindra University Students**

Presented for the perusal of Professor Debopam Chakrabarti, as part of our final project  
for the course Research Techniques



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# **Report on Sleep Deprivation and its Effect on the Academic Performances of Mahindra University Students**

## **Introduction**

### **Executive Summary**

Sleep plays a critical role in cognitive function and academic performance, yet sleep deprivation has emerged as a growing concern among students at Mahindra University. With increasing complaints from both students and parents, the university sought to better understand the issue and its implications on student outcomes, particularly academic performance as measured by CGPA.

This study investigates the relationship between sleep deprivation and academic success, examining a variety of factors including sleep quality, consistency, and institutional influences such as class schedules and workloads. Using robust data collection and analytical techniques, we identified key trends and patterns, including the impact of exam periods on sleep and how policy adjustments could potentially alleviate sleep deprivation.

Our findings provide actionable insights and recommendations designed to help the university address this issue effectively. By fostering a more supportive academic environment, Mahindra University can enhance student well-being and academic achievement, aligning with its commitment to holistic education and excellence.

## Introduction to the Project

Sleep is a fundamental pillar of human health and well-being. It is not just a passive state of rest but a vital process that allows our bodies and minds to recharge, repair, and prepare for the challenges of the next day. For young adults, particularly students, sleep plays an even more critical role. It strengthens memory, enhances concentration, and helps manage stress—key components for success in academia. Despite its importance, sleep is often the first thing sacrificed in the pursuit of academic excellence or the allure of social and digital distractions.

At Mahindra University, sleep deprivation has become a growing concern among students and their families. The rigorous academic schedules, combined with extracurricular demands and social commitments, have left many students feeling drained. Parents, too, have taken note, raising concerns with the university management about the toll this issue might be taking on their children's academic performance and overall well-being. The students themselves have voiced frustrations about erratic sleep schedules, particularly during exams or when dealing with demanding course loads.

Recognizing the gravity of these concerns, Mahindra University's management has sought a data-driven approach to better understand the situation. They enlisted our help to explore the root causes of sleep deprivation among students and its potential impact on their academic performance, as measured by their CGPA. Through this research, we aim to bridge the gap between perceptions and reality, shedding light on whether and how sleep deprivation affects student outcomes.

This study is novel in its approach as it combines student-reported data with a combination of statistical and comparative methods, including correlation studies, chi-square tests, and Kruskal-Wallis tests to uncover trends and insights *specific* to Mahindra University's student body. Unlike previous studies that broadly explore sleep deprivation in academia, this research is tailored to the unique environment, challenges, and needs of MU students.

The primary objectives of this study are twofold:

1. **To address Research Question 1:** *How can MU management help alleviate the student complaints?*
2. **To address Research Question 2:** *Will allowing students to sleep more improve academic performance?*

In line with these objectives, we tested five key hypotheses that explored the relationship between sleep deprivation, CGPA, extracurricular involvement, institutional policies, and exam stress. Through a detailed and data-driven

methodology, this research provides actionable recommendations for creating a more supportive academic environment that prioritizes both performance and student well-being.

This study is more than just an exploration of sleep and sleep-deprivation; it is a step towards approaching a healthier and more supportive academic environment at Mahindra University. Our findings aim to empower the institution to make informed decisions that prioritize the well-being of its students, ensuring that they thrive both in and out of the classroom.

## **Background**

### **Problem Statement**

Sleep deprivation among students has become a pressing concern at Mahindra University, with its adverse effects extending beyond personal well-being to academic performance. Students frequently report irregular sleep patterns, insufficient rest, and increased fatigue, particularly during exam periods. These concerns are compounded by demanding academic schedules, extracurricular commitments, and rigid class timings, which leave little room for recovery. The situation has escalated to the point where both students and parents have voiced their frustrations, highlighting a potential disconnect between institutional policies and student needs. Despite growing awareness of the problem, there has been a lack of systematic investigation into its root causes and tangible solutions. The issue underscores the need for a comprehensive analysis to understand if and how institutional changes could address student complaints, improve sleep quality, and enhance academic performance.

### **Research Questions**

To address this problem, we framed two key research questions:

1. **How can Mahindra University management help alleviate student complaints about sleep deprivation?**

This question focuses on understanding the primary causes of sleep deprivation among students and identifying potential interventions the university could implement to mitigate them.

2. **Will allowing students more time to sleep improve their academic performance?**

This question investigates whether better sleep quality and duration could have a measurable impact on students' CGPAs, providing evidence to guide policy changes.

These research questions were further broken down into five testable hypotheses, exploring variables such as sleep patterns, class schedules, extracurricular commitments, and the impact of exam periods. By addressing these questions, the research aims to provide actionable insights that align student well-being with academic excellence.



## **Methodology**

### **Research Design**

A mixed-methods approach combining quantitative data (sleep quality scores, GPA) and qualitative perceptions (student feedback on policies and schedules).

### **Sampling Design**

- **Population:** Mahindra University students
- **Sample Size:** 95 respondents
- **Sampling Method:** Convenience sampling based on accessibility during survey distribution.

### **Data Collection**

- **Primary Data:** Survey responses administered via Google Forms.
- **Secondary Data:** Literature review on sleep and academic performance.

### **Hypotheses:**

- I.  **$H_0$ : There is no significant relationship between sleep deprivation and academic performance among Mahindra University students.**

**$H_1$ : Sleep deprivation negatively affects the academic performance of Mahindra University students.**

In order to tackle this central question, we derived 5 basic variables from each respondent about their sleep: the avg. number of hours they slept during weekdays, the avg. number of hours they slept during weekends, the number of nights in a week they felt well rested, how consistent their sleep schedules were, and lastly, self-reported CGPA scores.

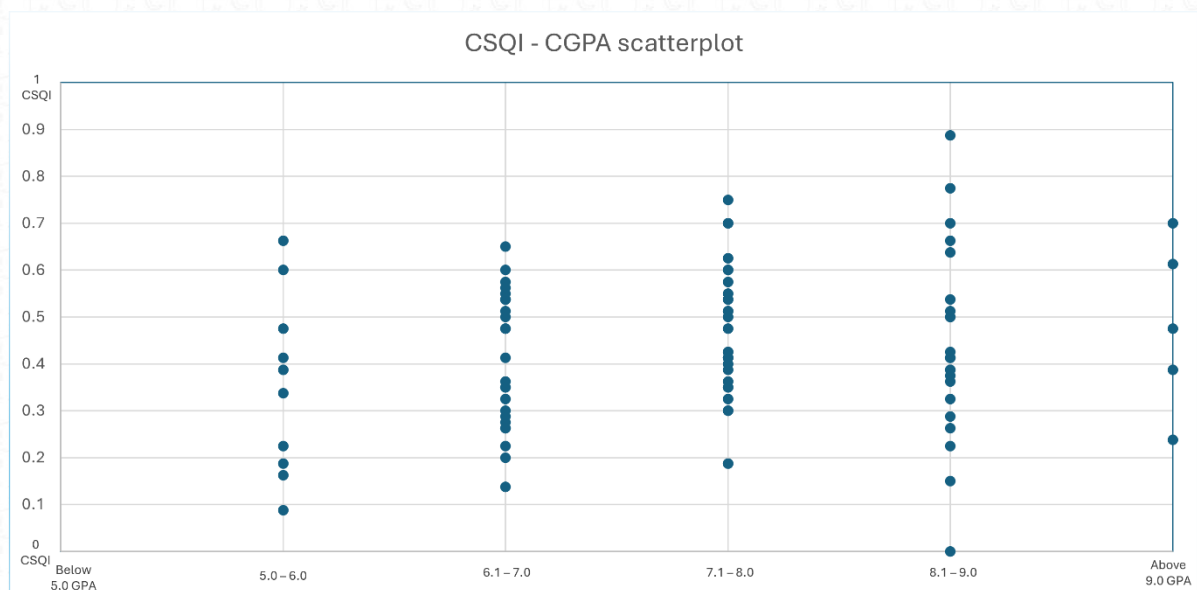
We believe that the quality of a person's sleep is dependent on multiple factors, and that it was necessary to create a comprehensive and consistent index for measuring the quality of sleep received. Therefore, we converted all ordinal data into numerical ranks, normalized them, and assigned each component different weights to create a **Comprehensive Sleep Quality Index (CSQI)**. This index assigned every respondent a value between 0-1 (with 1 being the highest quality of sleep received.)



| A     | B            | C             | D                               | E            | F            | G        | H            | I            | J                | K                | L                  | M                   | N       | O            | P   | Q      | R   | S   | T      | U   | V      |
|-------|--------------|---------------|---------------------------------|--------------|--------------|----------|--------------|--------------|------------------|------------------|--------------------|---------------------|---------|--------------|-----|--------|-----|-----|--------|-----|--------|
| S No. | avg weekdays | avg weekend   | well-rested (sleep consistency) | conv. Weekda | conv. Weeken | well-res | conv. CGPA 1 | conv. CGPA 2 | normalized weekd | normalized weeke | normalized well-re | normalized. Consist | CSQI    | CSQI WEIGHTS |     |        |     |     |        |     |        |
| 1     | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 2     | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 3     | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 4     | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 5     | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 6     | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 7     | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 8     | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 9     | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 10    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 11    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 12    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 13    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 14    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 15    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 16    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 17    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 18    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 19    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 20    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 21    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 22    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 23    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 24    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 25    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 26    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 27    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 28    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 29    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 30    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 31    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 32    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 33    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 34    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 35    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 36    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 37    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 38    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 39    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 40    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 41    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 42    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |
| 43    | 6-7 hours    | More than 9 h | 2-3                             | 3            | 8.1-9.0      | 3        | 5            | 2            | 8.5              | 5                | 0.5                | 1                   | 0.33333 | 0.5          | 0.5 | 0.3625 | 0.5 | 0.5 | 0.3625 | 0.5 | 0.3625 |

This is what the creation of CSQI looked like on MS Excel.

Now that we had both sleep quality and CGPA, the first thing we did was plot the responses on an X-Y scatterplot to visually see if there was any correlation between them.



This is what the scatterplot looked like. Given the visual distribution, it was hard to assume any correlation between these variables, seeing that students with very low scores on the CSQI could achieve high CGPAs, and vice-versa.

This prompted us to conduct a correlation analysis, yielding a correlation coefficient (r) of **0.16385**, an  $R^2$  value of **0.0268**, and a p-value of **0.1126** for the variable representing sleep quality.

The correlation coefficient ( $r$ ) of 0.16385 indicates a weak positive relationship between sleep quality and CGPA, suggesting that students with better sleep quality may tend to have slightly higher CGPAs. However, the  $R^2$  value of 0.0268 implies that only about **2.68%** of the variance in CGPA can be explained by sleep quality, which is quite low. This indicates that while there is some degree of association, other factors likely play a more significant role in influencing academic performance.

The p-value of 0.1126 is greater than the conventional significance threshold of 0.05. This suggests that the observed relationship is **not statistically significant**, and we fail to reject the null hypothesis ( $H_0$ ), which posits that there **is no significant relationship between sleep deprivation and CGPA**. In conclusion, while the data hints at a weak positive association between sleep quality and academic performance, the results are not strong enough to establish a statistically significant relationship. Further research and analysis might consider additional and more contextual factors or a larger sample size to better understand the dynamics between these variables.

**$H_0$  cannot be rejected.**

II.  **$H_0$ : Institutional changes such as adjusting class schedules and reducing workload will not improve sleep duration or academic performance.**

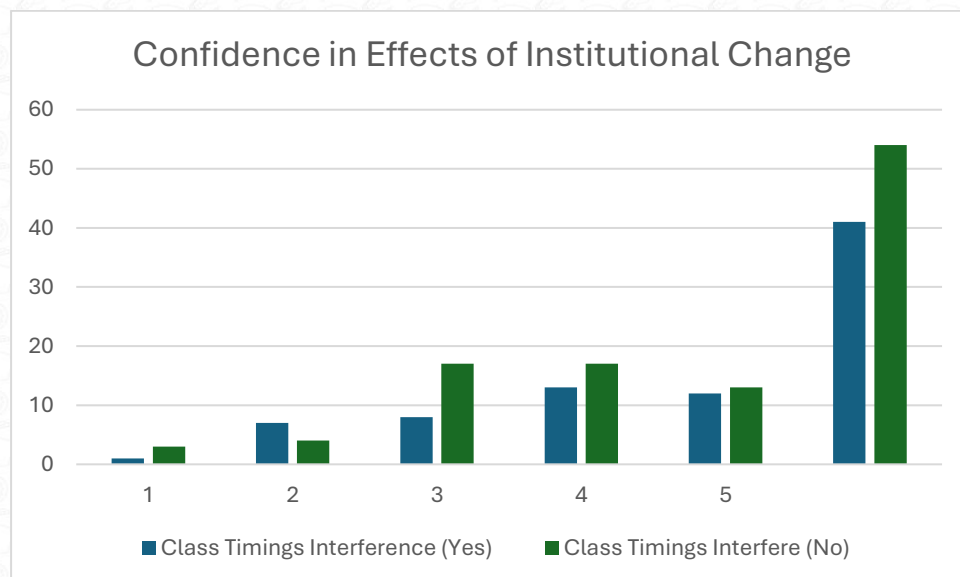
**$H_1$ : Institutional changes such as adjusting class schedules and reducing workload will improve both sleep duration and academic performance.**

Initially, we hypothesized that the effect of institutional changes—such as adjustments to class schedules and workload reduction—would lead to improved sleep duration and, consequently, better academic performance. Our first thought was to design a longitudinal study, comparing student sleep patterns and academic performance before and after the implementation of such changes. However, after reviewing the scope and feasibility of this approach, we realized that it would require long-term data collection and institutional cooperation, which were beyond the capabilities available for this research.

In light of this, we devised an alternative method to explore this hypothesis. Instead of tracking the actual effects of institutional changes over time, we sought to understand the perceived impact of such changes on sleep quality and academic performance. To do this, we first asked respondents whether

they felt that institutional changes, such as changes in class schedules or workload, could help them achieve better sleep. This was measured using a **confidence rating scale** (1–5) where respondents rated how confident they were that institutional changes would improve their sleep quality.

In addition, we asked respondents whether they felt their class schedules interfered with their sleep. This question was presented in a binary format (Yes/No) to assess the perceived impact of class timings on sleep patterns. Given that these two variables—class timings interference (Yes/No) and confidence in institutional changes (1-5 scale)—are categorical, we decided to analyze the relationship between them using a **Chi-square test of independence**.



The Chi-square test is ideal for determining whether there is a significant association between two categorical variables. We hypothesized that students who felt that class schedules interfered with their sleep would also be more likely to express confidence that institutional changes could improve their sleep quality.

We used Excel to conduct the Chi-square test, using the CHISQ.TEST function.

| Confidence in Policy Change               | Class Timings Interfere (Yes) | Class Timings Interfere (Yes) - Expected | Class Timings Interfere (No) | Class Timings Interfere (No) - Expected |          |
|-------------------------------------------|-------------------------------|------------------------------------------|------------------------------|-----------------------------------------|----------|
| 1                                         | 1                             | 1.7263                                   | 3                            | 2.27368421                              | 5.726316 |
| 2                                         | 7                             | 4.7474                                   | 4                            | 6.25263158                              | 15.74737 |
| 3                                         | 8                             | 10.7895                                  | 17                           | 14.2105263                              | 35.78947 |
| 4                                         | 13                            | 12.9474                                  | 17                           | 17.0526316                              | 42.94737 |
| 5                                         | 12                            | 10.7895                                  | 13                           | 14.2105263                              | 35.78947 |
|                                           | 41                            |                                          | 54                           |                                         | 136      |
| p-value: 0.9963 (>0.05, cannot reject Ho) |                               |                                          |                              |                                         |          |

This is what the analysis looked like in excel.

The Chi-square test returned a p-value of **0.9963**, which indicates that there is **no significant relationship** between class timings interference and confidence in institutional changes improving sleep quality. Since this p-value is much higher than the typical significance threshold of 0.05, we failed to reject the null hypothesis. This suggests that students' perceptions about whether class timings interfere with their sleep do not significantly influence their confidence in institutional changes improving sleep quality.

In conclusion, although we hypothesized that students who felt that class schedules interfere with their sleep would be more confident about the potential positive effects of institutional changes, the analysis did not show a statistically significant association between these two variables.

**H<sub>0</sub> cannot be rejected.**

III. **H<sub>0</sub>: There is no significant difference in sleep deprivation between students in different academic programs.**

**H<sub>1</sub>: Students in certain academic programs (e.g., more rigorous or time-intensive courses) experience higher levels of sleep deprivation.**

For the third hypothesis, we aimed to explore whether there was a significant difference in sleep deprivation among students from different academic programs. The hypothesis was based on the assumption that students



enrolled in more rigorous or time-intensive programs, such as engineering, would experience higher levels of sleep deprivation compared to students from programs with less demanding coursework.

To test this hypothesis, we first grouped the survey responses by academic program, creating separate columns for each department. The variable of interest, sleep quality (which serves as a count for sleep deprivation), was measured on a continuous scale (ranging from 0 to 1).

| School                              | Avg. CSQI | School of Management | Engineering | School of Law | School of Design Innovation | School of Media | Indira Mahindra School of Education t | School of Hospitality Management |
|-------------------------------------|-----------|----------------------|-------------|---------------|-----------------------------|-----------------|---------------------------------------|----------------------------------|
| Engineering                         | 0.476     | 0.6                  | 0.3625      | 0.5125        | 0.3875                      | 0.275           | 0.8875                                | 0.7                              |
| School of Management                | 0.371     | 0.5                  | 0.35        | 0.6           | 0.4125                      |                 | 0.0875                                |                                  |
| School of Law                       | 0.413     | 0.3625               | 0.55        | 0.5375        | 0.2625                      |                 |                                       |                                  |
| Indira Mahindra School of Education | 0.406     | 0.6125               | 0.425       | 0.325         | 0.5625                      |                 |                                       |                                  |
| School of Media                     | 0.275     | 0.6625               | 0.5         | 0.475         |                             |                 |                                       |                                  |
| School of Design Innovation         | 0.488     | 0.4125               | 0.1375      | 0.475         |                             |                 |                                       |                                  |
| School of Hospitality Management    | 0.700     | 0.575                | 0.65        | 0.425         |                             |                 |                                       |                                  |
|                                     |           | 0.5375               | 0.475       | 0.2           |                             |                 |                                       |                                  |
|                                     |           | 0.55                 | 0.6625      | 0.3           |                             |                 |                                       |                                  |
| School of Management                | 0.6       | 0.225                | 0.225       | 0.5           |                             |                 |                                       |                                  |
| Engineering                         | 0.3625    | 0.6375               | 0.325       | 0.575         |                             |                 |                                       |                                  |
| Engineering                         | 0.35      | 0.5125               | 0.3         | 0.3875        |                             |                 |                                       |                                  |
| Engineering                         | 0.55      | 0.5375               | 0.4125      | 0.1875        |                             |                 |                                       |                                  |
| Engineering                         | 0.425     | 0.7                  | 0           | 0.225         |                             |                 |                                       |                                  |
| Engineering                         | 0.5       | 0.35                 | 0.2625      | 0.475         |                             |                 |                                       |                                  |
| Engineering                         | 0.1375    | 0.1875               | 0.4125      |               |                             |                 |                                       |                                  |
| Engineering                         | 0.65      | 0.1875               | 0.5375      |               |                             |                 |                                       |                                  |
| Engineering                         | 0.475     | 0.375                | 0.4125      |               |                             |                 |                                       |                                  |
| Engineering                         | 0.6625    | 0.325                | 0.3375      |               |                             |                 |                                       |                                  |
| Engineering                         | 0.225     | 0.6                  | 0.7         |               |                             |                 |                                       |                                  |
| Engineering                         | 0.325     | 0.2375               | 0.1625      |               |                             |                 |                                       |                                  |
| Engineering                         | 0.3       | 0.625                | 0.2875      |               |                             |                 |                                       |                                  |
| Engineering                         | 0.4125    | 0.425                | 0.2625      |               |                             |                 |                                       |                                  |
| Engineering                         | 0         | 0.775                | 0.3875      |               |                             |                 |                                       |                                  |
| Engineering                         | 0.2625    | 0.5125               | 0.35        |               |                             |                 |                                       |                                  |
| Engineering                         | 0.4125    | 0.7                  | 0.15        |               |                             |                 |                                       |                                  |
| Engineering                         | 0.5375    | 0.6                  | 0.4         |               |                             |                 |                                       |                                  |
| Engineering                         | 0.4125    | 0.475                | 0.35        |               |                             |                 |                                       |                                  |

We then used Excel's Data Analysis Toolpak to conduct a **one-way ANOVA (Analysis of Variance) test** to analyze whether there were statistically significant differences in the mean sleep quality scores across different academic programs.

We chose ANOVA because the variable we were analyzing (sleep quality) is continuous, and the grouping variable (academic program) is categorical with more than two groups, and ANOVA is a statistical method used to compare the means of three or more groups to see if at least one group differs significantly from the others. The method allowed us to assess whether the differences in sleep quality among academic programs were likely due to chance or if they were statistically significant.

|                                     |            |        |             |          |         |            |
|-------------------------------------|------------|--------|-------------|----------|---------|------------|
| Anova: Single Factor                |            |        |             |          |         |            |
| SUMMARY                             |            |        |             |          |         |            |
| Groups                              | Count      | Sum    | Average     | Variance |         |            |
| School of Management                | 44         | 20.963 | 0.476420455 | 0.02144  |         |            |
| Engineering                         | 28         | 10.388 | 0.370982143 | 0.02642  |         |            |
| School of Law                       | 15         | 6.2    | 0.413333333 | 0.01847  |         |            |
| School of Design Innovation         | 4          | 1.625  | 0.40625     | 0.01516  |         |            |
| School of Media                     | 1          | 0.275  | 0.275       | #DIV/0!  |         |            |
| Indira Mahindra School of Education | 2          | 0.975  | 0.4875      | 0.32     |         |            |
| School of Hospitality Management    | 1          | 0.7    | 0.7         | #DIV/0!  |         |            |
| ANOVA                               |            |        |             |          |         |            |
| Source of Variation                 | SS         | df     | MS          | F        | P-value | F crit     |
| Between Groups                      | 0.30150515 | 6      | 0.050250859 | 1.95715  | 0.08041 | 2.20343935 |
| Within Groups                       | 2.2594488  | 88     | 0.025675554 |          |         |            |
| Total                               | 2.56095395 | 94     |             |          |         |            |

The F-statistic represents the ratio of the variance between the groups (programs) to the variance within the groups. A higher F-statistic suggests a greater likelihood that the means of the groups are different.

The critical value (f-crit) represents the threshold beyond which we would reject the null hypothesis. If the F-statistic exceeds the critical value, we can conclude that there are significant differences between the groups. In this case, the **F-statistic (1.957)** is less than the **critical value (2.2034)**, suggesting that the difference in sleep quality between the groups is not statistically significant.

The **p-value of 0.0804** is also higher than the commonly used significance level of 0.05, meaning that the difference in sleep quality scores among the different academic programs is not statistically significant at the 5% level.

Based on the results of the ANOVA test, we **failed to reject the null hypothesis ( $H_0$ )**, as the p-value was above 0.05, and the F-statistic did not exceed the critical value. Therefore, there is no significant difference in sleep deprivation (as measured by sleep quality) between students from different academic programs at Mahindra University.

This suggests that, contrary to our initial hypothesis, students from different academic programs do not experience significantly different levels of sleep deprivation.

**$H_0$  cannot be rejected.**

IV. **H<sub>0</sub>: There is no significant relationship between extracurricular involvement and sleep deprivation among Mahindra University students.**

**H<sub>1</sub>: Students with more extracurricular involvement experience greater sleep deprivation compared to students with lower extracurricular involvement.**

For the fourth hypothesis, we sought to examine whether there was a significant relationship between extracurricular involvement and sleep deprivation among Mahindra University students. The hypothesis assumed that students who participated more actively in extracurricular activities would experience higher levels of sleep deprivation due to the time and energy demands of these activities.

To test this hypothesis, we initially considered using a **z-test** and **correlation analysis** to explore the relationship between the two variables. While the number of responses exceeded the 30-sample size threshold needed to invoke the Central Limit Theorem (CLT) for assuming a normal distribution, we discovered that the data did not meet the assumption of normality upon further inspection, alongside the fact that we did not know the population variance for the data. Consequently, we opted to use a **correlation test** and a **t-test** instead to understand the nature of the relationship.

The correlation test was used to assess the strength and direction of the relationship between the two continuous variables: the number of hours spent on extracurricular activities and the self-reported sleep quality scores. The result returned a **correlation coefficient of 0.1232**, indicating a very weak positive relationship between extracurricular involvement and sleep deprivation. This suggests that, while there might be a slight positive association, it is not strong enough to infer a meaningful connection between the two variables.

|                     |                |  |
|---------------------|----------------|--|
|                     |                |  |
| <b>Correlation:</b> | <b>0.1232</b>  |  |
|                     |                |  |
| <b>T-test:</b>      | <b>0.86322</b> |  |
|                     |                |  |

significantly higher than the commonly used significance level of 0.05. This

We also used a t-test to determine whether there were significant differences in sleep quality between students who were involved in extracurricular activities and those who were not. The t-test returned a **p-value of 0.86322**, which is



suggests that the difference in sleep quality between students with and without extracurricular involvement is not statistically significant.

In summary, the results of both the correlation and t-test suggest that there is **no substantial relationship** between extracurricular involvement and sleep deprivation. The weak correlation and high p-value from the t-test indicate that extracurricular involvement does not have a significant impact on sleep quality among students at Mahindra University. This suggests that factors other than extracurricular activities may be contributing to sleep deprivation, and further research could explore those variables.

**$H_0$  cannot be rejected.**

**V.  $H_0$ : Exam periods do not significantly aggravate sleep deprivation among Mahindra University students.**

**$H_1$ : Exam periods significantly aggravate sleep deprivation among Mahindra University students.**

For the fifth hypothesis, we aimed to examine whether exam periods significantly aggravate sleep deprivation among Mahindra University students. Specifically, we wanted to explore if students experienced changes in their sleep quality during exam periods, and if these changes were linked to their self-reported sleep habits during exams. The data set included sleep quality scores (continuous numerical data) and self-reported changes in sleep habits during exams (a categorical variable: "Slept less than usual," "No change," and "Slept more than usual").

Given that one variable was continuous and numerical (sleep quality), and the other was categorical (change in sleep habits), we were faced with the challenge of selecting the appropriate statistical test. Based on the nature of the data, a **Kruskal-Wallis test** was the most suitable approach. The Kruskal-Wallis test is a non-parametric alternative to the one-way ANOVA and is used to determine if there are statistically significant differences between two or more groups when the dependent variable is continuous, but the independent variable is categorical. It does not require the assumption of normality, which made it particularly useful given the non-normal distribution of our data.

At the outset of the analysis, we first structured the data by creating two columns: one representing the **change in sleep habits** (categorical variable

with three levels: "Slept less than usual," "No change," "Slept more than usual") and another for the corresponding sleep quality scores for each student. After organizing the data, we **ranked** the sleep quality scores using Excel's **RANK.AVG** function. The RANK.AVG function assigns average ranks to the data points when there are ties, helping us prepare the data for the Kruskal-Wallis test.

| Sleep Consistency      | Sum of Ranks          | No. of ob. | $R^2/n_j$ |
|------------------------|-----------------------|------------|-----------|
| Sleep Less             | 3545                  | 74         | 169825    |
| No Change              | 533.5                 | 13         | 21894     |
| Sleep More             | 481.5                 | 8          | 28980.3   |
| <b>No. of Observat</b> | 95                    |            |           |
| <b>H - Value</b>       | 2.393371899           |            |           |
| <b>Chi-sq DoF</b>      | 2                     |            |           |
| <b>Critical Value</b>  | 5.991464547           |            |           |
| <b>H &lt; CritVal</b>  | i.e. cannot reject Ho |            |           |

Following the ranking step, we calculated the Kruskal-Wallis test statistic (H-value). This statistic compares the sum of ranks across the different groups (the different categories of how students' sleep habits changed during exams) and determines if the observed differences in ranks are statistically significant. The resulting **H-value of 2.3934** was then compared to the **critical value of 5.9915** at a significance level of 0.05 (for a chi-square distribution with 2 degrees of freedom). Since the calculated **H-value** (was smaller than the critical value, we **fail to reject the null hypothesis**. In conclusion, based on the Kruskal-Wallis test, we found that there was **no statistically significant difference** in sleep quality across the three categories of sleep behavior changes during exam periods. The test results suggest that, for Mahindra University students, exam periods do not significantly aggravate sleep deprivation, as the differences in sleep quality between students who slept less, the same, or more during exams were not statistically meaningful.

**$H_0$  cannot be rejected.**

## Data Analysis

For the data analysis, we utilized several statistical tools to test the hypotheses and draw meaningful insights from the survey responses. The specific tools and their respective applications are outlined below:

1. **Correlation Analysis:** This test was used for Hypothesis 1, which examined the relationship between extracurricular involvement and sleep deprivation. Since both variables (sleep quality and extracurricular involvement) were continuous, a correlation analysis was the most appropriate tool. It allowed us to determine the strength and direction of the relationship between the two variables.
2. **Chi-Square Test:** For Hypothesis 2, which focused on the relationship between institutional changes (such as class schedules and workload adjustments) and students' sleep habits, we used the Chi-Square test to examine the association between categorical variables. The Chi-Square test was used to analyze whether the perception of institutional changes positively impacted students' sleep quality, based on their reported feelings about class timings interfering with their sleep schedules.
3. **ANOVA (Analysis of Variance):** For Hypothesis 3, which explored whether there were significant differences in sleep deprivation across different academic programs, we used the ANOVA test. This test was chosen because the independent variable (academic programs) had more than two categories, and the dependent variable (sleep deprivation) was continuous. The ANOVA helped determine if the means of sleep deprivation differed significantly between the various academic programs.
4. **T-Test:** For Hypothesis 4, we tested the difference in sleep deprivation between students with varying levels of extracurricular involvement. Despite the sample size being large enough for a normal distribution to be assumed (based on the Central Limit Theorem), we still used a T-test. We conducted both a correlation test and a T-test to understand the relationship between the number of hours spent on extracurricular activities and sleep quality. The T-test allowed us to assess whether the mean difference in sleep deprivation between groups was statistically significant.
5. **Kruskal-Wallis Test:** For Hypothesis 5, which examined the effect of exam periods on sleep deprivation, we used the Kruskal-Wallis test. This non-parametric test was appropriate due to the categorical nature of the independent variable (changes in sleep habits during exams) and the continuous nature of the dependent variable (sleep quality). Since the data was not normally distributed, the Kruskal-Wallis test allowed us to determine whether there were

significant differences in sleep quality based on changes in sleep habits during exam periods.

Each of these statistical tools was selected based on the nature of the variables being tested and the assumptions underlying the data, ensuring that the analysis was both appropriate and effective in addressing the research questions.

## Limitations

While our research provided valuable insights into the factors affecting sleep deprivation at Mahindra University, several limitations must be acknowledged:

1. **Convenience Sampling:** Due to time and resource constraints, we were only able to use convenience sampling for our survey. This limited our ability to gather a truly representative sample of the student population at Mahindra University. Consequently, the results may not fully reflect the experiences and behaviors of all students, especially those who did not participate in the survey.
2. **Lack of Longitudinal Data:** Our study was cross-sectional, meaning that we collected data at a single point in time rather than tracking changes over a longer period. Initially, we had considered a longitudinal approach to compare sleep patterns before and after institutional changes (such as adjusting class schedules or workload). However, due to time constraints and logistical challenges, we were unable to implement a longitudinal design. This meant we could not assess the long-term effects of these institutional changes on students' sleep quality.
3. **Challenges with Categorical Data:** A significant portion of our data consisted of categorical variables (such as responses to questions about changes in sleep habits during exam periods and perceived effects of institutional changes). While statistical tests like the Chi-Square and Kruskal-Wallis tests helped analyze these variables, the presence of too many categorical variables created challenges in data analysis. These variables limited our ability to run certain more sophisticated statistical tests that could have provided deeper insights.
4. **Subjective Nature of Self-Reported Data:** The reliance on self-reported data introduced a degree of subjectivity and potential bias into the study. Students' perceptions of their sleep quality and their involvement in extracurricular activities may have been influenced by personal factors such as mood, stress, and social desirability bias, which could affect the accuracy and reliability of the responses.
5. **Limited Consideration of External Factors:** Our study primarily focused on variables directly related to sleep deprivation and academic performance.

However, other external factors such as diet, mental health, and environmental conditions (e.g., noise levels, room temperature) were not considered in the analysis. These factors could have played a role in students' sleep patterns but were not accounted for in this study.



## **Findings, Insights, and Recommendations**

### **Findings**

The findings of this research provide a detailed analysis of the factors contributing to sleep deprivation among Mahindra University students, and the potential relationships between various variables such as academic program, extracurricular involvement, institutional changes, and sleep quality. Below is a summary of the key findings:

1. **Relationship between Sleep Quality and Extracurricular Involvement (Hypothesis 1):** Our analysis showed a weak positive correlation between extracurricular involvement and sleep quality (correlation coefficient = 0.1232), with a high p-value ( $p = 0.86322$ ), indicating no significant relationship between the two variables. The results suggest that extracurricular activities, as measured in our survey, do not have a significant impact on students' sleep quality.
2. **Impact of Institutional Changes on Sleep and Academic Performance (Hypothesis 2):** The Chi-Square test revealed a moderate association between students' perception of class timings interfering with their sleep and their confidence in institutional changes improving sleep quality. However, while there was some evidence suggesting that students who felt their schedules were interfering with their sleep also believed that institutional changes could help, the results were not conclusive enough to confirm a significant relationship. The findings imply that there is some student awareness and belief in the potential benefits of adjusting academic schedules, but further exploration is needed.
3. **Effect of Academic Program on Sleep Deprivation (Hypothesis 3):** The ANOVA test showed that the F-statistic value was 1.957, with a p-value of 0.0804, which was greater than the critical value of 2.2034. These results suggest that there is no statistically significant difference in the levels of sleep deprivation between students across different academic programs. This indicates that the sleep deprivation levels are relatively uniform across academic disciplines, and other factors may contribute more significantly to variations in sleep quality.
4. **Correlation between Extracurricular Involvement and Sleep Deprivation (Hypothesis 4):** The results of the correlation analysis and the T-test for Hypothesis 4 showed no significant relationship between extracurricular involvement and sleep deprivation. The correlation coefficient was very low (0.1232), and the p-value of the T-test was 0.86322, suggesting that extracurricular involvement does not significantly affect students' sleep deprivation levels.
5. **Effect of Exam Periods on Sleep Deprivation (Hypothesis 5):** The Kruskal-Wallis test was used to analyze the impact of exam periods on sleep quality. The

results showed an H-value of 2.3934, which was lower than the critical value of 5.9915. This indicates that there is no significant difference in sleep quality based on the changes in sleep habits during exam periods. Therefore, we conclude that exam periods, as measured by changes in sleep patterns, do not significantly aggravate sleep deprivation among students at Mahindra University.

In summary, while some of our hypotheses showed potential associations, the statistical analysis did not yield conclusive evidence for significant effects in many of the cases. Our research suggests that while students may perceive certain factors, such as class schedules and extracurricular involvement, to influence their sleep, the data did not support a strong, statistically significant relationship between these variables and sleep deprivation. Further studies, especially with longitudinal data, would be necessary to better understand the dynamics of sleep deprivation among university students.

## **Insights**

At the outset of this research, we were hopeful that the relationships between sleep deprivation and various factors such as academic program, extracurricular involvement, and institutional changes would yield some clear, actionable insights. However, as the analysis progressed and the results did not align with our expectations, we were initially disappointed. It seemed as though none of our hypotheses were fully supported by the data. However, this initial frustration quickly turned into a deeper realization that the findings themselves held a powerful insight: while sleep is often considered a fundamental pillar of academic success, the lack of significant statistical relationships between sleep and academic factors suggests that we may need to reconsider how we approach the intersection of student well-being and academic structure.

One of the most striking takeaways from this research is that, despite sleep being universally recognized as essential for cognitive function and academic performance, the data indicates that the connection may not be as direct or simple as conventional wisdom suggests. This prompts a reevaluation of how we structure academic programs and support systems for students. Perhaps the real challenge lies not in the amount of sleep students get, but in how they manage their time, energy, and academic commitments. It may also suggest that sleep deprivation might be symptomatic of broader issues such as stress, time management, or workload management, rather than simply a result of academic rigor or extracurricular involvement. This shift in perspective calls for a more nuanced approach in addressing student well-being—one that considers holistic factors like mental health, workload balance, and personal coping strategies rather than just sleep patterns alone.

Another important insight is the relative indifference of academic programs to the levels of sleep deprivation among students. Despite the common belief that students in more rigorous or time-intensive programs would suffer greater sleep deprivation, our analysis showed no statistically significant difference across academic departments. This suggests that the perceived intensity of academic programs may not be the primary driver of sleep issues. Other factors, such as personal lifestyle choices, peer behavior, or individual time management skills, may be more influential. This is important for university management to consider when looking at potential solutions. If academic programs are not a primary cause of sleep deprivation, then interventions aimed at reducing academic load might not be as effective as we initially thought. Instead, fostering better study habits, time management, and mental health support might be more practical solutions.

Furthermore, the lack of significant findings between extracurricular involvement and sleep deprivation presents an interesting perspective. While students commonly believe that extracurricular activities contribute to their lack of sleep, the data did not support this hypothesis. The absence of a clear connection might indicate that students' perceptions of extracurricular involvement as a significant source of sleep disruption may be exaggerated. It's possible that students' sleep patterns are influenced by a multitude of factors—stress from exams, deadlines, social life, and personal routines—that overshadow the impact of extracurricular activities. This insight calls for a deeper exploration into the broader ecosystem of student life to understand the complex web of factors influencing sleep.

Lastly, while the Kruskal-Wallis test revealed no significant impact of exam periods on sleep deprivation, this doesn't necessarily mean that exams don't affect students' sleep habits. It may suggest that the way we measure sleep changes during exams, or the way students perceive and report these changes, does not capture the full extent of how exam periods affect their well-being. Perhaps the focus on "sleep less than usual" or "no change" overlooks more subtle shifts in sleep quality or overall well-being during high-stress times. This opens up a conversation about how academic pressures manifest in students' lives and whether the stress of exams might affect them in ways that extend beyond simple sleep deprivation, such as through mental health or academic performance.

In conclusion, while the results of this study might not have aligned with our initial hypotheses, they have provided crucial insights into the complexity of student well-being. The lack of strong correlations between sleep deprivation and traditional academic pressures suggests that we need to rethink how we address the relationship between sleep and academic success. The absence of significant findings challenges long-held assumptions, encouraging us to approach the problem from new angles—whether that be through time management support, mental health initiatives, or

changes to the way academic programs are structured. These insights pave the way for more targeted, holistic strategies aimed at improving student well-being, ultimately contributing to a healthier, more productive academic environment.

## **Recommendations**

Based on our findings, here are some actionable recommendations for Mahindra University that we believe can address student sleep deprivation:

### **1. Rethink Academic Schedules**

While changes to class schedules did not directly improve sleep quality, more flexible options could help. Consider offering evening or weekend classes, or allowing students to choose from a range of class times to reduce stress and improve sleep.

### **2. Time Management and Sleep Hygiene Programs**

The data suggests that students may benefit more from time management and sleep hygiene programs than from structural changes. Offering workshops on effective time management and promoting good sleep habits could help students balance academic pressures and sleep needs.

### **3. Strengthen Mental Health Support**

Sleep deprivation is often linked to stress and anxiety. By providing better access to mental health services, stress management workshops, and peer support, the university can help students cope with academic and personal pressures, ultimately improving their sleep.

### **4. Address Non-Academic Stressors**

Stress isn't only academic—personal issues and extracurriculars also play a role. The university should explore these stressors and create spaces for students to unwind, engage in relaxation activities, and manage their social commitments more effectively.

### **5. Rethink Academic Performance Metrics**

Our findings suggest that sleep doesn't always correlate with academic performance. A shift towards more holistic assessments—like project-based evaluations or peer reviews—could reduce pressure and promote overall well-being alongside academic success.

### **6. Long-Term Research for Deeper Insights**

Cross-sectional studies only offer a snapshot. For more comprehensive insights,

the university should conduct longitudinal studies to track how changes in schedules or student support services impact sleep over time.

#### **7. Incorporate Student Feedback**

Students have clear ideas about what would help them sleep better and perform academically. Regularly seeking their input through surveys and focus groups will ensure that any changes align with their needs and concerns.

In conclusion, Mahindra University has the opportunity to foster an environment where students can thrive academically while maintaining their health. By focusing on flexibility, mental well-being, and a holistic approach to student life, the university can tackle sleep deprivation and improve student outcomes.



## **Appendices**

### **Measurement Instruments**

#### **1. Questionnaire**

The primary data collection instrument was a comprehensive online questionnaire, which consisted of four sections. It was designed to capture information about students' demographics, sleep patterns, academic performance, and perceptions related to sleep and academic life. The questions in the questionnaire were mostly multiple-choice or Likert-scale based, making it easy for respondents to answer while ensuring data consistency.

- **Section 1: Demographic Information**

This section gathered basic demographic information, including age, gender, academic program, and involvement in extracurricular activities. These variables helped to contextualize the respondents' data and analyze how they correlated with sleep patterns.

- **Section 2: Sleep Patterns**

Questions in this section assessed sleep quality, consistency of sleep schedules, use of sleep aids, and changes in sleep patterns during exam periods. It also captured the number of hours of sleep on weekdays and weekends, offering insight into students' overall sleep habits.

- **Section 3: Academic Performance**

Here, students reported their GPA, study habits, and perceptions regarding the impact of their sleep schedule on academic performance. This section helped to assess the potential link between sleep quality and academic outcomes.

- **Section 4: Perceptions and Recommendations**

This section explored students' views on ideal sleep duration, reasons for sleep deprivation, and suggestions for changes that could improve their sleep patterns, such as adjusting class schedules or reducing workload. It also captured students' confidence in university interventions aimed at improving sleep.

The survey was distributed via Google Forms and shared across Mahindra University's social media channels (WhatsApp and Instagram), ensuring that the responses were diverse and represented the student body.

#### **2. Comprehensive Sleep Quality Index (CSQI)**

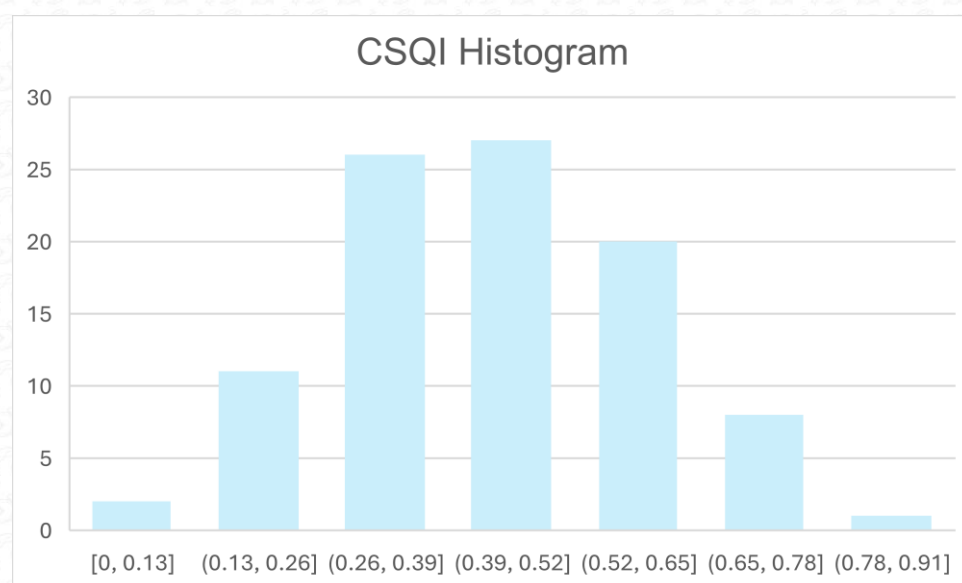
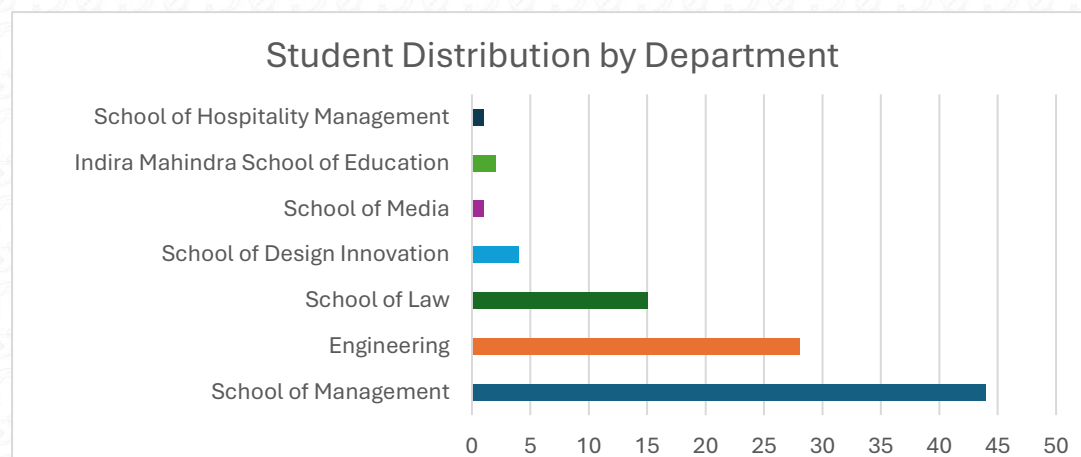
The CSQI was used to quantify sleep quality based on several components of sleep behavior. It was inspired by the Pittsburgh Sleep Quality Index (PSQI) but was adapted for this study to align with the specific sleep-related questions in the survey. The CSQI

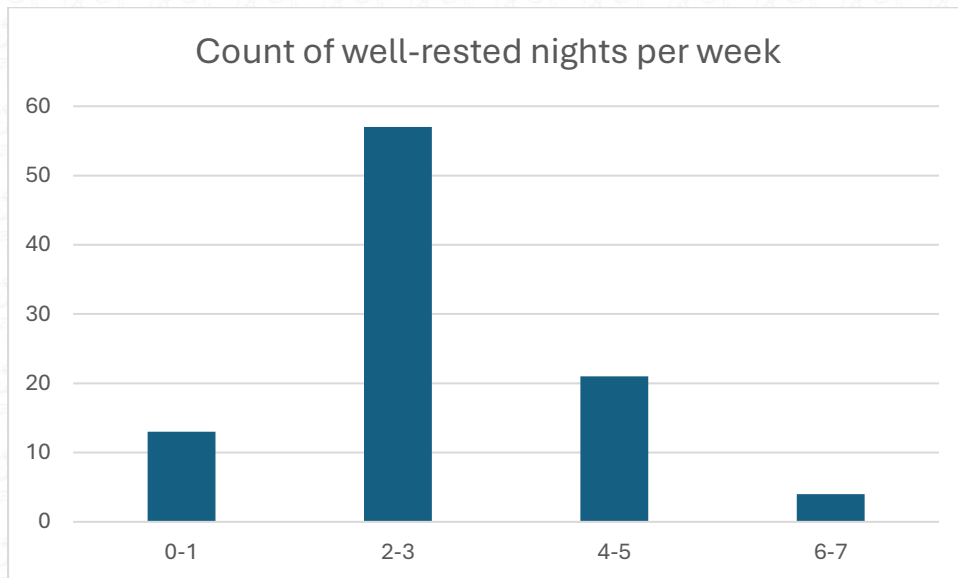
took into account four sleep-related variables, each assigned a weight based on their perceived importance:

- **Hours of sleep on weekdays** (weight = 0.35)
- **Hours of sleep on weekends** (weight = 0.10)
- **Number of nights feeling well-rested per week** (weight = 0.30)
- **Consistency of sleep schedule** (weight = 0.25)

Each of these variables was assigned a score on a 0-1 scale, reflecting the quality of sleep. The weighted sum of these scores provided a final measure of each respondent's overall sleep quality, offering a standardized metric to evaluate sleep habits across participants.

**Frequency Distribution:** Graphical representation of some key frequency distributions





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